Computer

Class 7 (English Medium)

Solved Exercises — Sindh Textbook Board

6 lessons — 123 questions

Fundamentals of ICT

[1-1] What can we do through 3D imaging/processing?

✓ (a) Show a hologram

(b) Move things from one factory to another

(c) Print a real picture

(d) Make a real decision about a sample

[1-2] What is possible through biometric verification?

(a) Identification through our account number

(b) Typing a password

✓ (c) Identification using face, finger prints and voice

(d) Identification through a phone number

[1-3] Which of the following is NOT called an emerging technology?

(a) Self-driving car

(b) Robotics

✓ (c) Banking

(d) Artificial intelligence

[1-4] Which input/output port is most commonly used to connect devices like a USB drive, mouse and printer to the computer?

(a) Serial port

(b) Port

✓ (c) USB port

(d) Parallel port

[1-5] What is the main purpose of peripheral devices?

✓ (a) To give the computer additional functionality

(b) To store files

(c) To send and receive data from external devices

(d) All of the above

[1-6] Devices like the mouse and keyboard are connected to the computer through _____.

Answer: to which devices are connected.

[1-7] The _____ port is used to connect a projector to the computer.

Answer: HDMI — the computer is connected to a monitor or projector through the HDMI

[1-8] _____ devices are devices that give input to the system unit and take output from it.

Answer: Peripheral — peripheral

[1-9] _____ are the connections on the front or back of the computer case that allow us to plug in I/O devices.

Answer: Ports — (the word-bank word is used again here) I/O devices

[1-10] The first non-commercial USB port was released in _____1994._____ .

Question: [1-11] Explain how modern technologies are changing this world.

Answer: Modern (emerging) technologies are changing the world by solving life's problems and bringing new ways: artificial intelligence teaches machines to think, learn and decide like humans; robotics lets machines sense their surroundings with sensors, decide on their own and work; virtual reality makes an artificial world feel real; biometrics identifies people by parts of the body; computer-assisted translation makes translation between languages easy; 3D holographic imaging shows a picture in the air; cloud computing gives storage, apps and software through the internet; open-source software allows everyone free use and modification.

Question: [1-12] Write in detail how the use of VR helps in education.

Answer: VR (Virtual Reality) is a technology that creates an artificial world which is not real but feels real, through special glasses or a headset. In education: students sitting in class can see historical places and experience things that are not present in front of them; training on dangerous equipment can be done safely; lessons become lively and interesting, which improves learning.

Question: [1-13] Explain in detail how the advancement of technology affects daily life.

Answer: The advancement of technology makes everything better, faster and cheaper: in communication there has been progress from the landline phone (1876) to the mobile phone (1990) and then the smartphone (after 2007) — now one phone runs the internet, photos, video, GPS, calls and learning/entertainment apps; in computing there has been progress from the large, expensive first-generation computers to the fast, cheap fourth/fifth-generation personal computers and mobiles; daily tasks (communication, shopping, learning, entertainment, work) have become faster and easier.

Question: [1-14] Write the difference between a printer and a scanner.

Answer: A printer is an output device: it converts the computer's information into a hard copy on paper, i.e. it prints text and pictures on paper (homework, reports, letters, pictures); a scanner is an input device: it copies things using light and converts paper into digital files, which can be saved and shared. In short: the printer goes from computer to paper, the scanner goes from paper to computer.

Question: [1-15] Why are peripheral devices necessary for the computer? Explain.

Answer: The computer (CPU) alone can do limited work; peripheral devices give it additional functionality: input devices (keyboard, mouse, webcam,

scanner) send data in; output devices (monitor, printer, speaker, projector) show the result; storage devices (USB drive, external hard drive, SSD) save data; network devices (modem, wireless access point, network card) connect the computer to the internet and networks. Without them the computer can neither take data in, nor show results, nor save files, nor connect.

Question: [1-A1] Activity Time: The teacher divides the class into two groups (A and B). Group A makes a paper-model robot and Group B shows a holographic image. Each group presents the work of its model.

Answer: Model completion: Group A — paper-model robot: a robot is a machine that senses its surroundings with sensors, decides on its own and works (examples: a robot arm in a factory, a vacuum-cleaner robot at home); Group B — holographic image: a 3D picture that appears in the air and can be seen from all sides, formed by rays of light (example: a hologram in a clothing shop).

Question: [1-A2] Activity Time: The teacher divides the class into groups and each group finds examples of common new technologies in their home and school (such as smartphone, computer, CCTV camera). Then each group prepares a list of those examples.

Answer: Model list: At home — smartphone, computer/laptop, smart TV, Wi-Fi router, smart watch; at school — computer lab, CCTV cameras, projector, smart board, printer.

Question: [1-A3] Activity Time: The teacher draws a Venn diagram on the board with two overlapping circles labelled "Modem" and "NIC Card" respectively. The students are asked to fill the circles with the functions of their devices. The students must find the differences in their functions and find similarities.

Answer: Model completion — Modem only: connects the computer to the

internet; converts the computer's data into phone-line signals and back; works between the computer and the Internet Service Provider (ISP). NIC card only: connects the computer to a network; it is a card fixed on the motherboard; it gives the computer an identity on the network. Similarities in both: both are network devices; both help connect the computer to the internet/network.

Digital Skills

[2-1] What is Microsoft Word?

(a) Spreadsheet

(b) Database management

(c) Presentation

✓ (d) Word processor

[2-2] The default extension of an MS Word file is _____.

(a) .txt

✓ (b) .docx

(c) .csv

(d) .ppt

[2-3] The extension of an MS PowerPoint file is _____.

(a) .doc

(b) .mp3

✓ (c) .ppt

(d) .mpt

[2-4] Which of the following is NOT a viewing format for a PowerPoint presentation?

(a) Slide view

(b) Outline view

(c) Slide show view

✓ (d) Presentation view

[2-5] Which keyboard shortcut is used to save a document immediately?

✓ (a) Ctrl + S

(b) Ctrl + D

(c) Alt + Shift + S

(d) Alt + S

[2-6] In which tab is the Header and Footer option found?

(a) Home

(b) Insert

(c) Layout

(d) Design

[2-7] In MS Word the ribbon is divided into different sections called _____.

Answer: has tabs such as Home, Insert, Design, Layout, etc.

[2-8] Mistakes in MS Word can be checked using the checked. facility.

[2-9] Use _____ in MS Word to make a copy of a file.

Answer: Save As — with Save As,

[2-10] The shortcut key for creating a new document in MS Word is _____.

Answer: blank document.

Question: [2-11] Write the uses of Microsoft Word.

Answer: With MS Word, documents can be created, edited, formatted and printed: letters, reports, assignments, projects, CVs, notes, books and forms. It has facilities for fonts, pictures, tables, headers/footers and spell check.

Question: [2-12] What are the basic functions of Microsoft Word?

Answer: Basic functions: creating a new document (Ctrl + N), typing and editing text, formatting (bold, italic, underline, font size and colour), spell check, saving (Ctrl + S), Save As, printing (Ctrl + P), inserting pictures/tables/pages, adding headers and footers, find and replace, and word count.

Question: [2-13] Explain the difference between Save and Save As.

Answer: Save (Ctrl + S) saves changes in the same file. Save As (Alt + F + A) saves the same file as a copy with a new name, new location or new format; the original file stays unchanged. Example: Save As is used to save a document as a PDF.

Question: [2-14] Write the purpose of a multimedia presentation.

Answer: The purpose of a multimedia presentation is to present information with text, pictures, sound, video and animation in such a way that the viewer understands easily, pays attention and remembers. Examples: a lesson presentation in class, a business report, introducing a project.

Question: [2-15] Define the terms: email address, password, contact; and write the uses of an email address.

Answer: Email address: a person's unique address on which email can be sent and received, e.g. user@example.com (username @ domain). Password: a secret code (letters + numbers + symbols, e.g. John@123) that keeps the email account safe; it must not be told to anyone. Contact: a person's saved information including their email address, so that they can be contacted easily. Uses of an email address: sending/receiving messages, creating accounts, recovering accounts, notifications, and sending documents.

Question: [2-16] Write the purpose of Inbox, Sent and Spam.

Answer: Inbox: the emails we have received appear in the Inbox. Sent: the emails we have sent are saved in Sent. Spam: unwanted/junk emails (adverts, fraud) that are automatically separated and filtered.

Question: [2-A1] Activity: Prepare a document about "My Best Friend". The teacher teaches the students to make a beautiful, colourful and well-arranged one-page document about their friend using the tools of the Insert, Design and Layout tabs. The following steps should be followed for

the practice: write the title; centre the title; use WordArt from the Insert tab; write a paragraph; write your friend's name; what do you like most about him/her?

Answer: Model completion: Title (WordArt, centred): "My Best Friend". Paragraph: My best friend's name is ___. He/she studies in class ___. What I like most about him/her is ___ (examples: he/she is truthful, helpful, studies together). Then make the document colourful and well-arranged (font, colour, picture) and save it with Ctrl + S.

Question: [2-A2] Activity: Make a PowerPoint presentation about My Dream Vacation. This activity is a complete 8-step project for a class setting, designed to make every student a PowerPoint expert in a 40-minute lab. Follow the steps below correctly and tick each step.

Answer: Model completion (8 steps): 1) Open PowerPoint: double-click the orange PowerPoint icon, choose a blank presentation. 2) Make 4 slides: Slide 1 title: "My Dream Vacation" (font size 44), below: "Ali Khan — Class 7-B — November 2025"; Slide 2 place: title "Maldives — Heaven on Earth" + 3 sentences (font size 28, 1200, tiny islands, water so clear you can see fish from the boat, 30°C every day); Slide 3 things to do: title "5 Famous Adventures" + 3 pictures, one line under each picture: snorkelling with turtles, night march under the stars, walking on the sandy shore; Slide 4 summary: title "I Must Go" + one big sentence "Blue water + kind people = perfect memories!". 3) Add transitions: choose the "Sea" theme from the Design tab; in the Transitions tab select all slides (Ctrl + A) → Fade → duration 01.00. 4) Add animations: Slide 2 sentences → Animation → Wipe; Slide 3 pictures → Zoom, Start: On Click. 5) Insert pictures: at least one picture per slide, resize (hold Shift for full shape). 6) Add footer/header: Insert → Header and Footer → type "Ali Khan — 7-B" in the footer → Apply to All. 7) Review the show: press F5, show each animation slowly with

clicks, press Esc to stop. 8) Save the file: File → Save As → Desktop.

Question: [2-CA1] Class Activity (i): Assign each group a task: such as making a birthday invitation, a party theme or a school timetable, educational details, etc. Help them research. Write a script using MS Word, design the script using different ribbons, so that a correct document can be made.

Answer: Model completion (birthday invitation example): Title (WordArt): "Birthday Invitation". Text: "My birthday is on ___ date at ___ time at ___ place. Please come!" Table: guest list. Design: colourful font, picture, border. Save (Ctrl + S).

Question: [2-CA2] Class Activity (ii): Ask the class to prepare the following PowerPoint presentation: give a test about geometric shapes, triangles, squares, rectangles, circles, etc.; animate a poem or quotation, animating each line with a different effect.

Answer: Model completion: Slide 1 title: "Geometric Shapes"; Slide 2: triangle, square, rectangle, circle — each shape's picture + name + number of sides; Slide 3: a poem or quotation, animate each line with a different animation (e.g. line 1 → Appear, line 2 → Fly In, line 3 → Zoom).

Question: [2-CA3] Class Activity (iii): Guide the students through the process of creating an email address using a famous mail service. Discuss the importance of choosing a professional username, name and password.

Answer: Model completion: Open Gmail (www.gmail.com) → click "Create account" → first and last name → choose a username (e.g. jane123@gmail.com) → make a strong password (letters + numbers + symbols) → confirm the password → enter a phone number → enter the verification code. Discuss why a professional username (your own name) and a strong password matter: the account stays safe and others can identify you easily.

Question: [2-CA4] Class Activity (iv): Instruct the students to practise sending an email to every student in their class. Instruct them to include a subject line, correct content and a signature. Encourage them to explore the To, Cc, Bcc fields and attach a file.

Answer: Model completion (email example): To: a friend's email address; Cc: the teacher's email address (for information); Bcc: (empty, for a hidden recipient); Subject: "About the class activity"; Content: "Assalam-o-Alaikum! I am ___. I am writing about the class activity..."; Signature: "Yours, ___, Class 7-B"; Attach a file → Send.

Algorithmic Thinking and Problem Solving

[3-1] Which of the following is NOT a part of computational thinking?

✓ (a) Estimation

(b) Saran

(c) Abstract thinking

(d) Logical thinking

[3-2] Which of the following is NOT a correct age?

(a) 12

✓ (b) -5

(c) 30

(d) 18

[3-3] Which of the following is an example of decomposition?

(a) Solving a puzzle all at once

✓ (b) Breaking a puzzle into pieces and edges

(c) Colouring a puzzle

(d) Looking at the puzzle's picture

[3-4] Which symbol is used to show the start or end of a flowchart?

(a) Rectangle

(b) Diamond

✓ (c) Oval

(d) Circle

[3-5] Why is it necessary to test an algorithm with correct and incorrect data?

(a) To make the algorithm difficult

(b) To guess what might happen

✓ (c) To make sure the algorithm produces correct results in all cases

(d) For fun

[3-6] The _____ symbol is used to represent a process in a flowchart.

Answer: Rectangle — the rectangle represents

[3-7] There are _____ types of loops.

Answer: Two — Finite Loop and Infinite Loop.

[3-8] _____ is the process of arranging a plan in a clear and step-by-step order.

Answer: Algorithm — an algorithm

[3-9] There can be ____many____ ways to solve a problem.

[3-10] The ____If-Else-If — this____ statement allows checking many conditions.

Question: [3-11] If we change the steps of the algorithm, how will the flowchart be used?

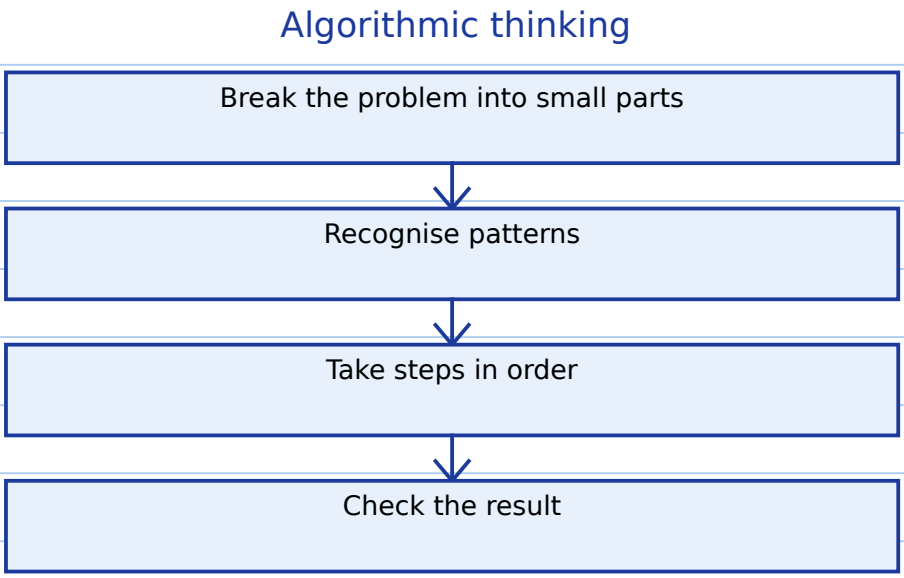
Answer: The flowchart must also be changed: every changed step of the algorithm must be changed in the flowchart with the same symbol, so that the flowchart and the algorithm stay identical. If the flowchart is not changed it becomes wrong, so after a change the flowchart must be checked (validated) again.

Question: [3-12] Explain how loops are different from conditional statements.

Answer: A conditional statement (if/else) chooses one path on the basis of a condition, and that step runs once. A loop (repetition) repeats the same instructions again and again while a condition stays true. In short: conditional statement = selection (once), loop = repetition (again and again).

Question: [3-13] Is algorithmic thinking important not only on computers but also in real life? Give an example.

Answer: Yes, algorithmic thinking is useful in daily life too: breaking a problem into small parts, recognising patterns, taking steps in order and checking the result. Example: making a cake — first gather the ingredients, then mix them, then bake, then taste (validation); or getting ready for school — clothes, bag, breakfast, route, in order.



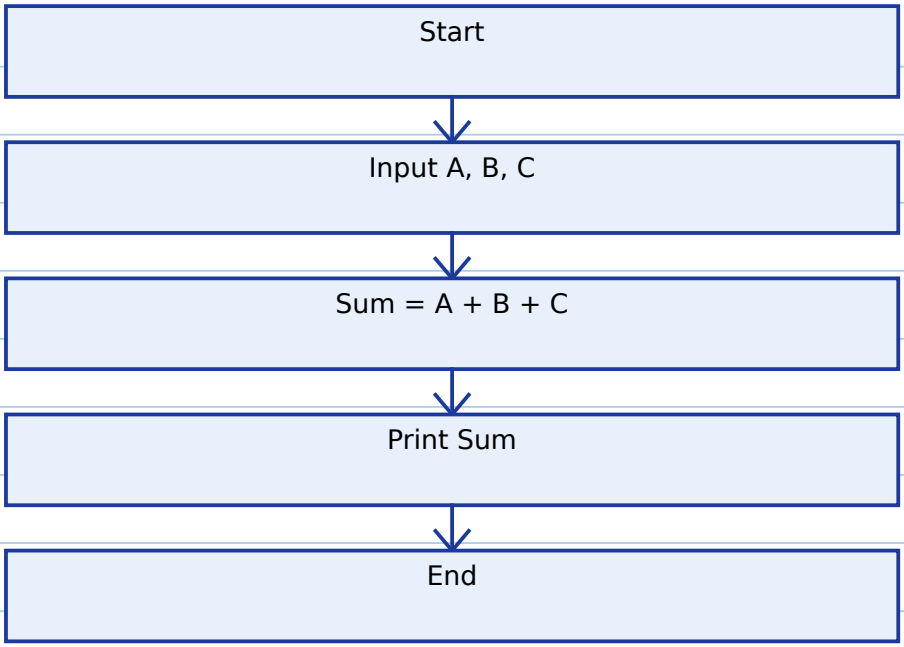
Question: [3-14] Why is it necessary to test an algorithm with different inputs?

Answer: So that it is verified that the algorithm works correctly with every kind of data: whether correct data gives the correct result, and whether incorrect data breaks the algorithm. Example: in checking age, both 12 (correct) and -5 (incorrect) are tried. If a wrong result is found, the steps are corrected.

Question: [3-15] Make a flowchart for a problem that shows the sum of three numbers.

Answer: Flowchart (in text): Start (oval) → Input A, B, C (parallelogram) → Sum = A + B + C (rectangle) → Print Sum (parallelogram) → End (oval).

Sum of three numbers — flowchart



Question: [3-16] Make a flowchart for a problem that only asks for a password. If the password is "SECRET", it says "Access granted to you" and otherwise it says "Access denied to you!".

Answer: Flowchart (in text): Start (oval) → Input Password (parallelogram) → Decision (diamond): Password = "SECRET"? → Yes → Print "Access granted to you" → End; No → Print "Access denied to you!" → End.

Question: [3-17] Make a flowchart for calculating the sum of the first 10 natural numbers.

Answer: Flowchart (in text): Start (oval) → N = 1, Sum = 0 (rectangle) → Decision (diamond): N ≤ 10? → Yes → Sum = Sum + N; N = N + 1 (rectangle) → back to decision; No → Print Sum (parallelogram) → End (oval).

Question: [3-18] Make a flowchart for a problem that keeps reaching numbers from 0 up to the entered number.

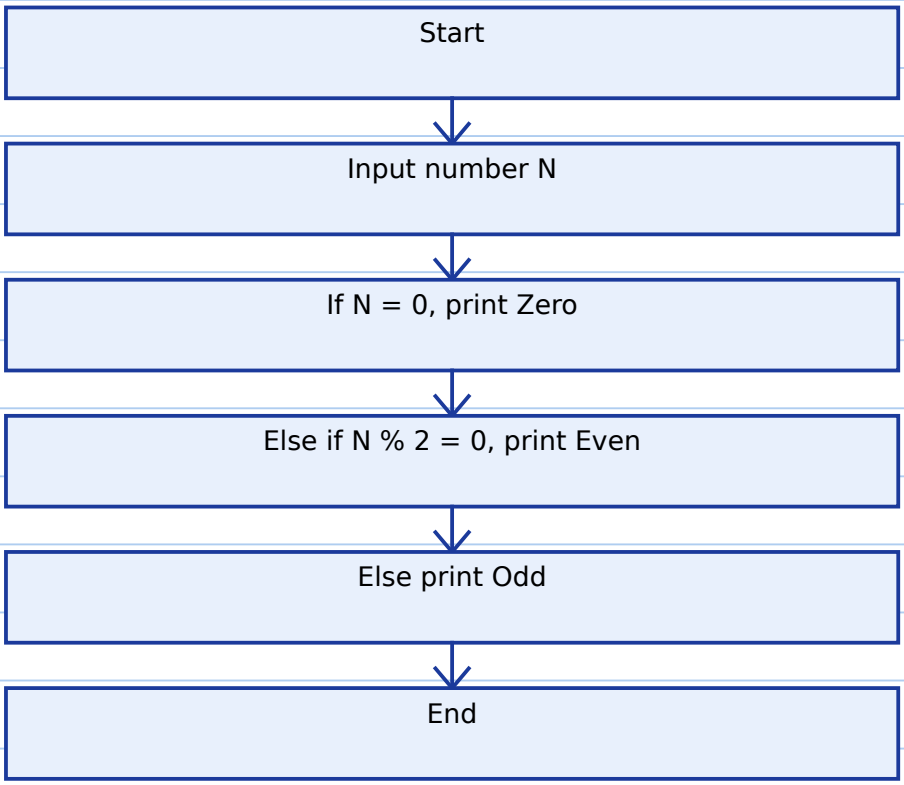
Answer: Flowchart (in text): Start (oval) → Input N (parallelogram) → Count = 0 (rectangle) → Decision (diamond): Count ≤ N? → Yes → Print Count;

Count = Count + 1 (rectangle) → back to decision; No → End (oval).

Question: [3-19] Write an algorithm that takes an input of a number and then checks whether the number is Even, Odd, or Zero.

Answer: Step 1: Start. Step 2: Input number N. Step 3: If N = 0, print "Zero". Step 4: Else if N % 2 = 0, print "Even". Step 5: Else print "Odd". Step 6: End.

Even, Odd or Zero — algorithm



Question: [3-20] Write an algorithm that asks the price of a book and then applies a 10 percent discount on the original price and shows the final price after the book.

Answer: Step 1: Start. Step 2: Input the book price P. Step 3: Discount = P × 0.10. Step 4: Final Price = P − Discount. Step 5: Print Final Price. Step 6: End.

Question: [3-21] (Hint) The 10 percent discount is calculated by multiplying the original price by 0.10; then subtracting the resulting price

from the original price reduces it.

Answer: Explanation: Discount = Original Price × 0.10; Final Price = Original Price − Discount. Example: price 1000 rupees → discount = 1000 × 0.10 = 100 rupees → final price = 1000 − 100 = 900 rupees.

Question: [3-22] Zawiyar is at the bookshop to buy a book. If the price of a book is more than 500 rupees, or if Zawiyar is a student, Zawiyar will get a 20 percent discount. Otherwise, no discount will be given in any case.

Answer: Algorithm: Step 1: Start. Step 2: Input price P and student status. Step 3: If P > 500 OR Student = Yes, then Discount = P × 0.20; Final Price = P − Discount; print Final Price. Step 4: Else print P (no discount). Step 5: End.

Question: [3-23] Write an algorithm that takes a name as input and prints it 10 times on the screen.

Answer: Step 1: Start. Step 2: Input Name. Step 3: Set Count = 1. Step 4: If Count ≤ 10, print Name; Count = Count + 1; repeat Step 4. Step 5: End.

Question: [3-A1] Activity Time: Think about a difficult problem for arranging elections for class monitor. Use computational thinking to solve this real-world problem. You must make sure that the election process is true, fair, simple and well-organised.

Answer: Model solution (with computational thinking): Decomposition — break the problem into parts: nomination, voting, counting, announcement. Abstraction — ignore unnecessary information (favourite colour, etc.); keep only the voter list, candidates' names and votes. Logical thinking — clear order: announcement → nomination → voting → counting → announcement. Decision making — rules: one vote per student, secret ballot, re-voting on a tie. Validation — test with sample votes that the counting is correct.

Question: [3-A2] Activity Time: Analyse the given examples carefully and

think why the "unclear instructions" given in the third column are unclear. (Example: Check whether a number is Even or Odd — clear instructions: 1. Start. 2. Input number N. 3. If N % 2 = 0 show "Even". 4. Else show "Odd". 5. End. Unclear instructions: Take a number. Decide whether it is equal or odd. Show the result.)

Answer: Analysis: The unclear instructions are unclear because they do not say HOW to decide — no clear rule (N % 2 = 0) is given for checking "equal/even"; there is no start/end; it is not clear in which case to show the result. Clear instructions give every step, condition and result with the correct number and order.

Question: [3-CA1] Class Activity (i): Consider the following algorithm for calculating total tax: a customer buys two items (mouse and keyboard), input the price of the mouse (P1) and the price of the keyboard (P2); tax on the mouse = P1 × 0.10; tax on the keyboard = P2 × 0.10; Total Tax = Tax1 + Tax2; print Total Tax; end. Make a flowchart to represent the given algorithm.

Answer: Flowchart (in text): Start (oval) → Input mouse price P1 (parallelogram) → Input keyboard price P2 (parallelogram) → Tax1 = P1 × 0.10 (rectangle) → Tax2 = P2 × 0.10 (rectangle) → Total Tax = Tax1 + Tax2 (rectangle) → Print Total Tax (parallelogram) → End (oval).

Question: [3-CA2] Class Activity (ii): Identify which of the following tasks represent a loop — "Finite" or "Infinite". (1) Keep watering the plant until the soil becomes wet. (2) Test a game until the health of the players under guidance reaches Zero. (3) A computer program keeps checking new emails continuously. (4) A traffic light keeps changing colours again and again during the day. (5) A robot waters 3 plants one after another.

Answer: (1) Finite — when the soil becomes wet the loop stops. (2) Finite — when health reaches Zero the game ends. (3) Infinite — the program

never stops on its own, it keeps checking. (4) Finite — it repeats during the day and then stops (off at night). (5) Finite — exactly 3 plants, then end.

Question: [3-CA3] Class Activity (iii): Identify which of the following tasks represent a condition — If / Or / If-Else / If-Else-If-Else. (1) If the password is correct, then log in; otherwise show an error message. (2) If the user clicks "Submit", then save the form. (3) If the temperature is more than 40°C, then show a heat warning; otherwise, if the temperature is less than 10°C, then show a cold warning; otherwise show normal. (4) If the file exists, open it. (5) If the fan is run it is hot. (6) If the payment succeeds, show the receipt; otherwise show an error. (7) If the traffic light is red, then stop; else go. (8) If the mobile battery is low, show a battery warning. (9) If the student is late, mark absent; otherwise mark present. (10) If Wi-Fi is connected, go online.

Answer: (1) If-Else (two branches: log in or error). (2) If (one condition, one action). (3) If-Else-If-Else (three cases: hot, cold, normal). (4) If. (5) If — note: this sentence is printed unclearly in the book. (6) If-Else. (7) If-Else. (8) If. (9) If-Else. (10) If. Note: none of the ten given examples clearly matches an "Or" (OR) condition.

Programming

[4-1] ASCII stands for:

(a) American Scientific Character for Information Interchange

✓ (b) American Standard Code for Information Interchange

(c) American Simple Clause for Information Interchange

(d) American Structural Character for Information Interchange

[4-2] In Scratch the building blocks are divided into many categories; the block below (move 10 steps) belongs to which category?

(a) Events

(b) Operators

✓ (c) Motion

(d) Control

[4-3] Which of the following is NOT a binary number?

(a) 1111

(b) 101

✓ (c) 11E

(d) 000

[4-4] With reference to the picture below (Scratch script: when green flag clicked, go to x: 0 y: 0, repeat until x position > -10, move 10 steps, stop all), which of the options below can move the sprite to (10,0)?

(a) Click the green flag

(b) Click the cat

(c) stop all

✓ (d) move 10 steps

[4-5] _____ is the central coordinate of the XY plane.

✓ (a) X=0, Y=0

(b) X=0, Y=180

(c) X=-240, Y=0

(d) X=0, Y=-240

Question: [4-6] How is information stored on the computer?

Answer: The computer stores information in binary (0 and 1): with encoding, letters/pictures/sound are converted into binary code (e.g. with ASCII, A = 01000001) and that code is stored in memory/storage.

Question: [4-7] Differentiate between the binary and decimal number systems.

Answer: Binary system: base 2, only two digits (0 and 1), the computer's system. Decimal system: base 10, ten digits (0 to 9), humans' everyday system.

Question: [4-8] How many types of block palettes are there in Scratch programming?

Answer: 9 types: Motion, Looks, Sound, Events, Control, Sensing, Operators, Variables, My Blocks.

Question: [4-9] Explain events in Scratch and give some examples.

Answer: An event is the trigger that starts a script. Examples: when green flag clicked, when sprite clicked, when a key is pressed (when space key pressed), when the backdrop switches, when a message is received. Without events a program does not start.

Question: [4-10] Define the following terms: (i) Sprite (ii) Stage (iii) Costume (iv) Variable.

Answer: (i) Sprite: the character or object in Scratch that moves, speaks or interacts with others (e.g. the cat). (ii) Stage: the background where all the activity happens. (iii) Costume: a different shape/appearance of the sprite — changing the costume makes the sprite look different. (iv) Variable: a container/box that stores a value (such as score, name, lives, time) which

can change.

[4-11] The base value in the binary number system is 2. .

[4-12] The binary equivalent of the decimal number 10 is = 8 + 2 = 1010. .

[4-13] _____ is the decimal value of the binary equivalent 10101.

Answer: 21 — 10101 = 16 + 0 + 4 + 0 + 1 = 21.

[4-14] The computer understands only binary (0 and 1). .

Question: [4-A1] Activity Time: Convert the following decimals into binary numbers: (a) 54 (b) 21 (c) 35 (d) 47.

Answer: (a) 54 = 110110 (32 + 16 + 4 + 2). (b) 21 = 10101 (16 + 4 + 1). (c) 35 = 100011 (32 + 2 + 1). (d) 47 = 101111 (32 + 8 + 4 + 2 + 1).

Question: [4-A2] Activity Time: Convert the following binary numbers into decimal numbers: (a) 1001 (b) 1100 (c) 1000 (d) 1010.

Answer: (a) 1001 = 9 (8 + 1). (b) 1100 = 12 (8 + 4). (c) 1000 = 8. (d) 1010 = 10 (8 + 2).

Question: [4-A3] Activity Time: How to change the sprite's position? The easy way to change the sprite's position in Scratch is to use the Motion Block. This program moves the sprite to a fixed place when the Green Flag is clicked. (1) From the Events category, check the when green flag clicked block. (2) From the Motion category, check the go to x: 0 y: 0 block. It will position the sprite at the centre of the stage. (3) From the Motion category, check the glide 1 secs to x: 0 y: 0 block to set the X and Y values to the desired position.

Answer: Model solution (Scratch script): when green flag clicked → go to x: 0 y: 0 → glide 1 secs to x: 100 y: 50 (desired position, for example). Result: clicking the green flag brings the sprite to the centre and then it glides to

(100, 50).

Question: [4-A4] Activity Time: Allow the user to move the sprite (cat, parrot) using the Arrow Keys when the program starts, or let the sprite move automatically across the whole screen without any intervention when the program starts.

Answer: Model solution (two ways): (1) With arrow keys: when green flag clicked → forever → if up arrow key pressed then change y by 10; if down arrow key pressed then change y by -10; if right arrow key pressed then change x by 10; if left arrow key pressed then change x by -10. (2) Automatic: when green flag clicked → go to x: 0 y: 0 → forever → move 10 steps → if on edge, bounce.

Question: [4-CA1] Class Activity (1): Make a book. (The word printed in the book is "kutub".)

Answer: Model completion (Scratch story-book): make the sprites the characters (e.g. cat and dog), make the backdrops the book's pages (Backdrop 1, 2, 3), put the characters' speech (say block) and movement (move/glide) on each page, and use the next backdrop block to change pages.

Question: [4-CA2] Class Activity (2): Give the player score points in some situations, such as when they can touch another sprite that is moving in some way, or when the playing player clicks on the sprite.

Answer: Model solution (Scratch script): make a variable "Score". when green flag clicked → set Score to 0 → forever → if touching [other sprite] then change Score by 1. Separate script: when this sprite clicked → change Score by 1.

Digital Citizenship

[5-1] Which of the following is an example of appropriate online behaviour?

(a) Sharing someone's private pictures

(b) Spreading wrong information

✓ (c) Using polite language in messages

(d) Ignoring cyberbullying

[5-2] What does the term 'netiquette' refer to?

(a) The study of computer viruses

✓ (b) Appropriate and respectful behaviour on the internet

(c) Illegal online activities

(d) Rules for online shopping

[5-3] Which is a positive effect of using social media?

(a) Online scams

✓ (b) Making global friends and learning

(c) Cyberbullying

(d) Screen-time habit

[5-4] Which is a proper way to protect the computer from viruses?

(a) Disabling the firewall

(b) Sharing passwords

(c) Opening unknown email attachments

✓ (d) Using antivirus software

[5-5] What does digital prejudice mean?

(a) Being fair to all opinions

(b) Spreading rumours

✓ (c) Treating people unfairly online

(d) Sharing someone's personal data

[5-6] Respecting another person's opinion is an example of _____ behaviour online.

Answer: appropriate

[5-7] _____ software helps protect the device from viruses.

Answer: Antivirus — antivirus

[5-8] Sending a _____ message to a classmate on game chat is an example of cyberbullying.

Answer: hurtful/bad

[5-9] Spreading _____ about someone is an inappropriate act.

Answer: Rumours — spreading rumours

[5-10] Downloading _____ software is an example of misuse of computer resources.

Answer: Illegal — downloading illegal (pirated)

Question: [5-11] What is netiquette?

Answer: Netiquette is the rules of good manners and etiquette in the online world: speaking politely, respecting others' opinions, not sharing personal information, not spreading rumours, being mindful of time, and not hurting others.

Question: [5-12] What is cyberbullying?

Answer: Cyberbullying is using technology (mobile, internet, social media) to repeatedly hurt, embarrass or frighten someone: bad messages, rumours, sharing private pictures, removing from a group, impersonation.

Question: [5-13] Name one positive effect of using social media.

Answer: Making global friends and learning — with social media, friends can be made around the world, new things can be learned, contact with family stays, and knowledge can be shared.

Question: [5-14] What is digital prejudice?

Answer: Digital prejudice is treating people unfairly online on the basis of their race, religion, gender, language or identity — e.g. removing someone from a group because of their background, or valuing their opinion less.

Question: [5-15] What is a computer threat? Also give five examples of computer threats.

Answer: A computer threat is anything that can harm the computer, the information stored in it, or the user. Five examples: (1) Virus (2) Malware (3) Hacking (4) Phishing (5) Spam.

Question: [5-CA1] Class Activity 1: Read each action below. Tick (✓) the box that shows whether the action is appropriate or inappropriate. Actions: (1) Helping a friend who forgot their homework. (2) Reporting cyberbullying or any harmful behaviour to a trusted adult. (3) Visiting safe websites and asking an adult before downloading apps or files. (4) Ignoring a friend who fell down and got hurt. (5) Saying "please" and "thank you" with recognition. (6) Spreading rumours about someone.

Answer: (1) Appropriate ✓. (2) Appropriate ✓. (3) Appropriate ✓. (4) Inappropriate ✗. (5) Appropriate ✓. (6) Inappropriate ✗.

Question: [5-CA2] Class Activity 2: Read each situation below. Tick (✓) the box that shows whether the situation is an example of cyberbullying or not cyberbullying. Situations: (1) Your friend sends a funny video of a cat. (2) Someone sends a bad message to your classmate on game chat. (3) You mistakenly send a text message to the wrong person via WhatsApp. (4) A person posts another student's embarrassing picture online without permission. (5) Someone removes you from an online group where your friends are. (6) Your teacher sends a link to an educational video.

Answer: (1) Not cyberbullying ✓ (funny video, hurts no one). (2) Cyberbullying ✓ (bad message). (3) Not cyberbullying ✓ (a mistake, no intent). (4) Cyberbullying ✓ (embarrassing picture without permission). (5)

Cyberbullying ✓ (removal from a group = exclusion). (6) Not cyberbullying ✓ (educational link).

Entrepreneurship (Business Idea / New Venture)

[6-1] A person who starts a new business is called?

(a) Customer

(b) Innovator

✓ (c) Entrepreneur

(d) Manager

[6-2] The art of folding paper into beautiful shapes?

(a) Design thinking

(b) Business

✓ (c) Origami

(d) Innovation

[6-3] How many special goals has the United Nations made?

(a) 10

(b) 15

✓ (c) 17

(d) 20

[6-4] The stage where a product or idea is tested whether it works or not?

(a) Idea

(b) Reflect

✓ (c) Test

(d) Define

[6-5] A model or sample of a product is called?

(a) Service

(b) Design

✓ (c) Prototype

(d) Plan

[6-6] Entrepreneurship means that a person identifies an opportunity and starts a new _____ to solve a problem.

Answer: business

[6-7] Innovation helps people change _____ ways of thinking and try new ways.

Answer: old

[6-8] Design thinking is a _____ method for solving real-life problems.

Answer: creative, step-by-step

[6-9] The stage where the product is improved on the basis of feedback is called _____.

Answer: improvement/trying again.

[6-10] A _____ or service is work or help that a person provides to others to meet their needs.

Answer: Product — a product

Question: [6-11] What is problem identification?

Answer: Problem identification means finding and recognising a problem or need in daily life or the community — it is the first stage of design thinking and business. Examples: litter in school, water shortage, long queues.

Question: [6-12] What is the name of the stage where a real business starts selling finished products to customers?

Answer: Implementation — in the Implementation stage the business starts selling the finished product to real customers in the market.

Question: [6-13] What is the design thinking process?

Answer: The design thinking process (8 stages): (1) Identify — find the problem. (2) Empathize — feel others' problem. (3) Define — state the problem clearly. (4) Ideate — think of new ideas. (5) Prototype — make a model. (6) Test — test it. (7) Reflect — learn from feedback. (8) Iterate —

improve and try again. Then Implementation.

Question: [6-14] Also give five examples of problem-solving in daily life.

Answer: (1) Broken tap — fit a new tap. (2) Forgetting homework — make a checklist. (3) Dark room — fit a lamp/light. (4) Litter in school — dustbins and a cleanliness campaign. (5) Long queue — a token system.

Question: [6-15] What is Sustainable Development Goal number one (1) aimed at ending?

Answer: Ending poverty (No Poverty) — Goal 1 aims to end every form of poverty in the world.

Question: [6-A1] Class Activity 1: Understanding Product vs Service. Objective: To understand the difference between a product and a service. Activity: 1. Solve a problem: The teacher will divide the class into groups and give each group a problem card, for example "People need help getting groceries delivered home." 2. Think about the solution: Each group will think of a solution that is a product (a grocery cart) or a solution that is a service (a delivery service). 3. Share with the class: Each group will explain its product or service, and state how a product is different from a service.

Answer: Model completion: Problem — people need help getting groceries delivered home. Product solution: a grocery cart (a thing that can be bought, seen/touched). Service solution: a home delivery service (work that a person does, cannot be seen). Difference: a product is a thing, a service is work/help.

Question: [6-A2] Group Activity 2: "Problem Finders" Scavenger Hunt. Objective: To observe a real place, and find things that cause trouble or difficulty for people in doing daily work comfortably. Steps: 1. Make teams: The class will be divided into small groups. 2. Get helping materials: Each team will be given a notebook and pencil by the teacher. 3. Search: Each

group will go to a place like the classroom, library, or playground and see which things do not work correctly. 4. Write: Problems will be written at each place, such as "Finding a book in the library is difficult" or "The chairs are too small or uncomfortable." 5. Group discussion: Each group will share the problems it found with the class. The teacher will ask: "Why is this a problem for people?" 6. Conclusion: The teacher will explain that this activity helps students learn about understanding people's feelings and the importance of solving problems.

Answer: Model completion (observation examples): (1) Finding a book in the library is difficult — solution: labels/list on books. (2) The chairs are too small or uncomfortable — solution: right-sized chairs. (3) The dustbin in the classroom is far away — solution: keep a bin nearby. (4) The water tap on the playground is broken — solution: repair the tap. Discussion answer: these are problems for people because they make daily work difficult and slow.

Question: [1-CA1] Class Activity: The teacher takes the students to the library so that they can observe the I/O ports of the computer there. Fill in the table below: Name of I/O port | Name of peripheral device.

Answer: Model completed table: USB port — mouse, keyboard, printer, pen drive; HDMI port — monitor, projector; Ethernet port — network cable (router); Audio jack — headphones, speaker; Power port — charger.